

SQL Analytical Tasks

Step-by-Step Guide

Tank54 Group | Oracle HR Schema

What is in this book?

This guide has two parts. **Part 1** shows 5 examples. Each example has a task, step-by-step explanation, and the SQL query. **Part 2** has 15 practice tasks for you to solve.

Topics: SELECT, WHERE, ORDER BY, Aggregate Functions, GROUP BY, HAVING, Subqueries

Oracle HR Schema - EMPLOYEES Table

We use the Oracle HR database. For all tasks in this guide, we use only the **EMPLOYEES** table. This table has information about every employee in the company.

Column	Type	Example	Description
EMPLOYEE_ID	Number	100	Unique ID for each employee
FIRST_NAME	Text	Steven	First name of the employee
LAST_NAME	Text	King	Last name of the employee
EMAIL	Text	SKING	Email of the employee
PHONE_NUMBER	Text	515.123.4567	Phone number
HIRE_DATE	Date	17 - JUN - 03	Date when the employee started
JOB_ID	Text	AD_PRES	Job role code
SALARY	Number	24000	Monthly salary
COMMISSION_PCT	Number	0.3	Commission percent (some jobs)
MANAGER_ID	Number	(null)	ID of the employee's manager
DEPARTMENT_ID	Number	90	ID of the department

Quick Reminder - Key Concepts

Concept	What it does	Example
COUNT (*)	Counts rows	SELECT COUNT(*) FROM employees
SUM(column)	Adds all values	SELECT SUM(salary) FROM employees
AVG(column)	Calculates average	SELECT AVG(salary) FROM employees
MIN(column)	Finds smallest value	SELECT MIN(salary) FROM employees
MAX(column)	Finds largest value	SELECT MAX(salary) FROM employees
GROUP BY	Groups rows by a column	GROUP BY department_id
HAVING	Filters after GROUP BY	HAVING COUNT(*) > 5
WHERE	Filters rows before grouping	WHERE salary > 10000
ORDER BY	Sorts the result	ORDER BY salary DESC
Subquery	Query inside a query	WHERE salary > (SELECT AVG(salary) FROM employees)

PART 1: Examples

In this part, we show 5 examples. Each example has 3 parts: the task, the steps, and the SQL query. Read each step carefully. Try to understand the logic before you look at the SQL code. All examples use only the EMPLOYEES table. No JOINS are used.

Example 1: Employee Count by Department

Task:

Count how many employees work in each department. Show department ID and the count. Show only departments with more than 5 employees.

Step-by-Step:

Step 1: What do we want to count? Employees in each department. So we need COUNT(*) to count rows.

Step 2: We need to count by department, so we use GROUP BY department_id. This puts employees into groups by their department.

Step 3: We only want departments with more than 5 employees. Because 5 is about the COUNT result, we use HAVING (not WHERE).

Step 4: Remember: WHERE filters rows before grouping. HAVING filters groups after grouping.

SQL Query:

```
SELECT department_id,
       COUNT(*) AS employee_count
FROM employees
GROUP BY department_id
HAVING COUNT(*) > 5
ORDER BY employee_count DESC;
```

Example 2: Average, Min and Max Salary

Task:

Find the average, minimum, and maximum salary for each job. Show job ID and all three values. Order by average salary from high to low.

Step-by-Step:

Step 1: We need three aggregate functions: AVG(salary), MIN(salary), and MAX(salary). All three work on the SALARY column.

Step 2: We want these values for each job, so we use GROUP BY job_id.

Step 3: We use ROUND() to make the average salary show only 2 decimal places. This makes the result cleaner.

Step 4: We order by the average salary in DESC order (high to low). We can use the alias "avg_sal" in ORDER BY.

SQL Query:

```
SELECT job_id,
       ROUND(AVG(salary), 2) AS avg_sal,
       MIN(salary) AS min_sal,
       MAX(salary) AS max_sal
FROM employees
GROUP BY job_id
ORDER BY avg_sal DESC;
```

Example 3: Employees Earning Above Average

Task:

Find all employees whose salary is higher than the average salary of all employees. Show name, salary, and how much more they earn than the average.

Step-by-Step:

Step 1: First, we need the average salary. A simple subquery gives this: SELECT AVG(salary) FROM employees.

Step 2: We use this subquery in the WHERE clause. Each employee's salary must be > the subquery result.

Step 3: To show how much more they earn, we subtract the subquery from the salary: salary - (SELECT AVG(salary) FROM employees).

Step 4: We concatenate first name and last name with || to show the full name.

SQL Query:

```
SELECT first_name || ' ' || last_name AS employee_name,
       salary,
       salary - (SELECT AVG(salary) FROM employees) AS above_avg
FROM employees
WHERE salary > (SELECT AVG(salary) FROM employees)
```

```
ORDER BY salary DESC;
```

Example 4: Salary Ranges - Counting Groups

Task:

Put employees into 3 salary groups: Low (salary < 5000), Medium (5000 to 10000), High (> 10000). Count how many employees are in each group.

Step-by-Step:

Step 1: We need to create a group label for each employee based on their salary. We use CASE WHEN for this.

Step 2: CASE WHEN salary < 5000 THEN 'Low' WHEN salary <= 10000 THEN 'Medium' ELSE 'High' END. The conditions check from top to bottom.

Step 3: We wrap the CASE expression in an outer query and use GROUP BY to count employees in each salary group.

Step 4: COUNT(*) counts the rows in each group. We order the result so "High" comes first.

SQL Query:

```
SELECT salary_group,
       COUNT(*) AS employee_count
FROM (
    SELECT CASE
        WHEN salary < 5000 THEN 'Low'
        WHEN salary <= 10000 THEN 'Medium'
        ELSE 'High'
    END AS salary_group
    FROM employees
) GROUP BY salary_group
ORDER BY employee_count DESC;
```

Example 5: Departments Where Avg Salary is Below Company Average

Task:

Find departments where the average salary is lower than the company average. Show department ID, average salary, and the company average.

Step-by-Step:

Step 1: We need the company average salary. A subquery gives this: SELECT AVG(salary) FROM employees.

Step 2: We need the average salary per department. We use GROUP BY department_id and AVG(salary).

Step 3: To compare a group result with another value, we use HAVING with a subquery. We cannot use WHERE here because AVG is an aggregate function.

Step 4: We show both the department average and the company average side by side for easy comparison.

SQL Query:

```
SELECT department_id,
       ROUND(AVG(salary), 2) AS dept_avg,
       ROUND((SELECT AVG(salary) FROM employees), 2) AS company_avg
FROM employees
WHERE department_id IS NOT NULL
GROUP BY department_id
HAVING AVG(salary) < (SELECT AVG(salary) FROM employees)
ORDER BY dept_avg;
```

PART 2: Practice Tasks

Now it is your turn! Solve these 15 tasks. Write your SQL queries on paper or in a database tool. Use only the EMPLOYEES table. No JOINs are needed. Good luck!

Task 1	Total Salary Cost of the Company	SUM
Find the total salary cost of the whole company. This means: add up all employee salaries. Show the total as one number.		
Task 2	How Many Employees in the Company?	COUNT
Count the total number of employees in the company. Show the count as one number.		
Task 3	Salary Statistics	AVG + MIN + MAX
Find the average, minimum, and maximum salary in the whole company. Show all three values in one result.		
Task 4	Employee Count by Department	GROUP BY + COUNT
Count how many employees are in each department. Show department ID and the count. Order by count from high to low.		
Task 5	Average Salary by Job	GROUP BY + AVG + ORDER BY
Find the average salary for each job ID. Show job ID and average salary (round to 2 decimals). Order by average salary from high to low.		
Task 6	Departments with More Than 10 Employees	GROUP BY + HAVING + COUNT
Find departments that have more than 10 employees. Show department ID and the count. Order by count from high to low.		
Task 7	High-Earning Jobs	GROUP BY + HAVING + AVG
Find jobs where the average salary is more than 10000. Show job ID and average salary. Order by average salary from high to low.		
Task 8	Employees Earning Above Average	Subquery + WHERE
Find all employees who earn more than the company average salary. Show full name and salary. Use a subquery for the average.		
Task 9	Employees Earning Below Average	Subquery + WHERE + ORDER BY
Find all employees who earn less than the company average salary. Show full name and salary. Order by salary from low to high.		
Task 10	Salary Difference From Average	Subquery + Arithmetic + WHERE
For each employee, show their name, salary, and the difference from the company average (salary minus average). Show only employees who earn more than the average.		
Task 11	Commission Employees	WHERE IS NOT NULL + COUNT
Find all employees who have a commission (commission_pct is not null). Count how many such employees exist. Show the count.		
Task 12	Salary Ranges - Low, Medium, High	CASE WHEN + GROUP BY + COUNT
Group employees into 3 salary ranges: Low (less than 5000), Medium (5000 to 10000), High (more than 10000). Count how many employees are in each range. Use CASE WHEN.		
Task 13	Employees per Manager	GROUP BY + HAVING + COUNT
Count how many employees each manager has. Show manager ID and the count. Show only managers who have more than 5 employees. Order by count from high to low.		
Task 14	Highest Salary per Department	GROUP BY + MAX + ORDER BY
Find the highest salary in each department. Show department ID and the maximum salary. Order by maximum salary from high to low.		

Task 15**Departments with Highest and Lowest Salary Gap***GROUP BY + MAX - MIN + HAVING*

For each department, find the difference between the highest and lowest salary. Show department ID and the salary gap. Show only departments where the gap is more than 8000. Order by gap from big to small.